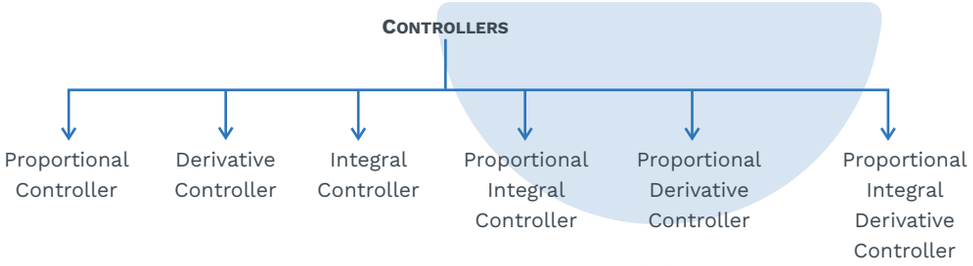
8 Controllers & Compensators

INTRODUCTION

An automatic control system is used to maintain its output within desirable limits using control action. The deviation detected by error detector is used as an actuating signal for control action through a controller whereas the compensators are used to improve the performance i.e. transient and steady state response characteristic during run time. The changes can be in the form of adjustment of forward path gain (or) feedback gain (or) inserting a compensating device i.e. compensators in the control system.

CONTROLLERS

There are various types of controllers (or) control actions which are required for improving transient and steady state response of a control system.



(1) Proportional (P) Controller

In case of proportional controller, the actuating signal for the control action in a control system is proportional to the error signal.

The error signal being the difference between the reference input signal and the feedback signal.

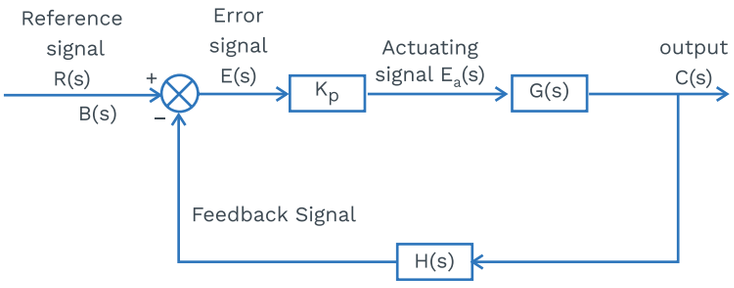


Figure-8.1. Proportional Control Action

Hence,

$$e_a(t) = K_p e(t)$$

Where, K_p = proportional constant

Taking Laplace transform, we get-

$$E_a(s) = K_p E(s)$$

Feedback path transfer function is given by, H(s) = 1
The overall transfer function of a second order closed loop system with proportional controller is given by-

$$\frac{C(s)}{R(s)} = \frac{K\omega_n^2}{s^2 + 2\xi\omega_n s + K\omega_n^2}$$

The characteristic equation for overall transfer function is given by-

$$s^2 + 2\xi\omega_n s + K\omega_n^2 = 0$$

Now, the damping ratio and natural frequency of the system using proportional controller is-

$$\omega_n' = \omega_n\sqrt{K}$$

$$\xi' = \frac{\xi}{\sqrt{K}}$$

The forward path transfer function and feedback path transfer function is given by-

$$G(s) = \frac{K\omega_n^2}{s(s + 2\xi\omega_n)},\ H(s) = 1$$

Hence, the transfer function relating E(s) and R(s) is given by-

$$\frac{E(s)}{R(s)} = \frac{1}{1 + G(s)H(s)}$$

$$\Rightarrow \quad \frac{E(s)}{R(s)} = \frac{1}{1 + \frac{K\omega_n^2}{s(s + 2\xi\omega_n)}}$$

$$\Rightarrow \quad E(s) = \frac{R(s).s(s + 2\xi\omega_n)}{s^2 + 2\xi\omega_n s + K\omega_n^2}$$

For the ramp input,

$$R(s) = \frac{1}{s^2}$$

$$\Rightarrow \quad E(s) = \frac{1}{s^2}\frac{s(s + 2\xi\omega_n)}{s^2 + 2\xi\omega_n s + K\omega_n^2}$$

Hence, the steady state error is given by-

$$e_{ss} = \lim_{s\to 0} sE(s)$$

$$\Rightarrow \quad e_{ss} = \frac{2\xi}{K\omega_n}$$

Gray Matter Alert!!!

The proportional band of any controller is defined as the range of the control input needed to cause the full change to the controller output.

The control system needs to be underdamped for quick response but in case of sluggish overdamped response of control system, to fasten the process, forward path gain has to be increased which reduces the steady state error and at the same time maximum overshoot increases.

(2) Derivative (D) Controller (Rate mode)

In case of derivative controller, the actuating quantity for the control action consists of the derivative of the error signal.

Hence,

$$e_a(t) = T_d \frac{de(t)}{dt}$$

where,

$T_d = K_d$ = Derivative gain constant or rate constant

By taking Laplace transform-

$E_a(s) = sT_d\, E(s)$

For first order unity feedback control system-

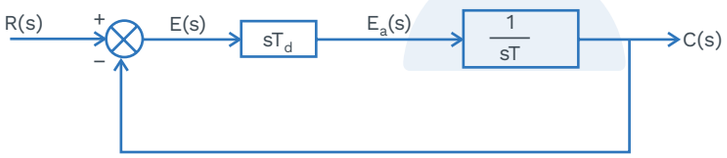


Figure 8.2. First Order System with Derivative Controller

The forward path transfer function G(s) and feedback path transfer function H(s) is given by-

$$G(s) = \frac{T_d}{T} \text{ and } H(s) = 1$$

The transfer function relating E(s) and R(s) is given by-

$$\frac{E(s)}{R(s)} = \frac{1}{1 + G(s)H(s)}$$

$$\Rightarrow \quad \frac{E(s)}{R(s)} = \frac{T}{T + T_d}$$

$$\Rightarrow \quad E(s) = R(s)\, \frac{T}{T + T_d}$$

Previous Year's Questions

A first order system with a proportional controller exhibits an offset to step input. The offset can be reduced by:

(a) Increasing the gain

(b) Adding integral mode

(c) Adding derivative gain

(d) Decreasing the gain

Ans. (b)

For unit step input-

$$R(s) = \frac{1}{s}$$

$$E(s) = \frac{1}{s}\frac{T}{(T + T_d)}$$

The steady state error is given by-

$$e_{ss} = \lim_{s \to 0} sE(s)$$

$$\Rightarrow \quad e_{ss} = \frac{T}{T + T_d}$$

Hence, the derivative controller improves the transient response of the system.

(3) Integral (I) Controller

In case of integral controller, the actuating quantity for the controller action consists of the integral of the error signal.

Hence,

$$e_a(t) = K_i \int e(t)dt$$

where,

K_i = Integral gain constant

By taking Laplace transform, we get-

$$E_a(s) = \frac{K_i E(s)}{s}$$

Rack your Brain

What is the Reset time of the controller?

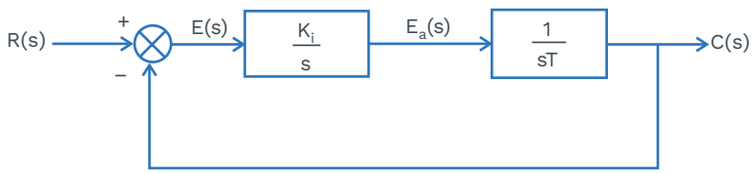


Figure-8.3. First order system with integral controller

The overall transfer function of a closed loop first order control system using integral controller is given by-

$$\frac{C(s)}{R(s)} = \frac{K_i / T}{s^2 + K_i / T}$$

For unit Ramp input-

$$R(s) = \frac{1}{s^2}$$

$$\Rightarrow C(s) = \frac{1}{s^2}\frac{K_i / T}{s^2 + \frac{K_i}{T}}$$

The forward path transfer function G(s) and feedback path transfer function H(s) is given by:

$$G(s) = \frac{K_i}{Ts^2} \text{ and } H(s) = 1$$

The transfer function relating E(s) and R(s) is given by-

$$\frac{E(s)}{R(s)} = \frac{1}{1 + G(s)H(s)}$$

$$\Rightarrow \quad \frac{E(s)}{R(s)} = \frac{1}{1 + \frac{K_i}{Ts^2}}$$

For Ramp input-

$$E(s) = \frac{1}{s^2}\frac{s^2}{s^2 + \frac{K_i}{T}}$$

$$\Rightarrow \quad E(s) = \frac{1}{s^2 + \frac{K_i}{T}}$$

The steady state error is given by-

$$e_{ss} = \lim_{s \to 0} sE(s)$$

$$\Rightarrow \quad e_{ss} = 0$$

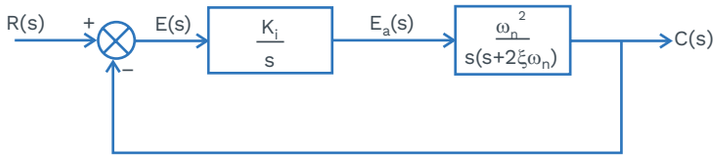


Figure-8.4. Second Order system with integral controller

The overall transfer function of a closed loop second order control system using integral controller is given by-

$$\frac{C(s)}{R(s)} = \frac{K_i\omega_n^2}{s^3 + 2\xi\omega_n s^2 + K_i\omega_n^2}$$

The forward path transfer function G(s) and feedback path transfer function H(s) is given by-

Previous Year's Questions

Introduction of integral action in the forward path of a unity feedback system results in a:

(a) Marginally stability
(b) System with no steady state error
(c) System with increased stability margin
(d) System with better speed of response

Ans. (b)

$$G(s) = \frac{K_i\omega_n^2}{s^2(s+2\xi\omega_n)} \text{ and } H(s) = 1$$

The transfer function relating E(s) and R(s) is given by-

$$\frac{E(s)}{R(s)} = \frac{1}{1+G(s)H(s)}$$

$$\Rightarrow \quad \frac{E(s)}{R(s)} = \frac{1}{1+\frac{K_i\omega_n^2}{s^2(s+2\xi\omega_n)}}$$

$$\Rightarrow \quad E(s) = R(s)\frac{s^2(s+2\xi\omega_n)}{s^3+2\xi\omega_n s^2+K_i\omega_n^2}$$

For unit Ramp input-

$$R(s) = \frac{1}{s^2}$$

$$\Rightarrow \quad E(s) = \frac{1}{s^2}\frac{s^2(s+2\xi\omega_n)}{s^3+2\xi\omega_n s+K_i\omega_n^2}$$

The steady state error is given by-

$$e_{ss} = \lim_{s\to 0} sE(s) \quad \Rightarrow \quad e_{ss} = 0$$

Gray Matter Alert!!!

The first order and second order control system along with integral controller is marginally stable and unstable with zero steady state error for unit Ramp input.

(4) Proportional Derivative (PD) Controller:

In case of proportional derivative controller, the actuating signal consists of the proportional error signal added with the derivative of the error signal.

Hence,

$$e_a(t) = e(t) + T_d\frac{de(t)}{dt}$$

where, T_d = constant

By Laplace transform-

$$E_a(s) = E(s) + sT_dE(s).$$

For second order unity feedback control system-

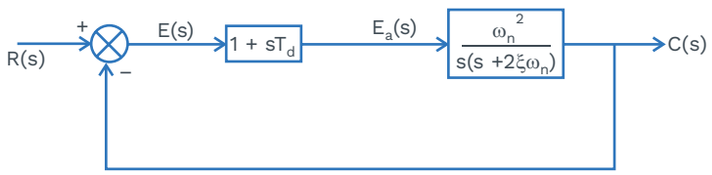


Figure-8.5. Proportional Derivative Controller

The overall transfer function of a closed loop second order control system using proportional derivative controller is given by-

$$\frac{C(s)}{R(s)} = \frac{(1+sT_d)\omega_n^2}{s^2 + (2\xi\omega_n + \omega_n^2 T_d)s + \omega_n^2}$$

The characteristics equation for overall transfer function is given by-

$$s^2 + (2\xi\omega_n + \omega_n^2 T_d)s + \omega_n^2 = 0$$

Now, the damping ratio of the system using derivative controller is-

$$\xi' = \frac{2\xi\omega_n + \omega_n^2 T_d}{2\omega_n}$$

$$\Rightarrow \quad \xi' = \xi + \frac{\omega_n T_d}{2}$$

Hence, by using proportional derivative controller, the effective damping is increased, and hence maximum overshoot is reduced.
The overall transfer function can be written as-

$$\frac{C(s)}{R(s)} = \omega_n^2 T_d \frac{\left(s + \frac{1}{T_d}\right)}{s^2 + 2\xi'\,\omega_n s + \omega_n^2}$$

Hence, the forward path transfer function G(s) is-

$$G(s) = (1+sT_d)\frac{\omega_n^2}{s(s+2\xi\omega_n)}$$

And, feedback path transfer function is-

$$H(s) = 1$$

The transfer function relating E(s) and R(s) is given by-

$$\frac{E(s)}{R(s)} = \frac{1}{1+G(s)H(s)}$$

$$\Rightarrow \quad \frac{E(s)}{R(s)} = \frac{s(s+2\xi\omega_n)}{[s^2 + (2\xi\omega_n + \omega_n^2 T_d)s + \omega_n^2]}$$

For unit ramp input,

$$R(s) = \frac{1}{s^2}$$

$$\Rightarrow \quad E(s) = \frac{1}{s^2} \cdot \frac{s(s+2\xi\omega_n)}{[s^2 + (2\xi\omega_n + \omega_n^2 T_d)s + \omega_n^2]}$$

The steady state error for unit ramp input is-

$$e_{ss} = \lim_{s\to 0} sE(s)$$

$$\Rightarrow \quad e_{ss} = \frac{2\xi}{\omega_n}$$

Here, the steady state error and natural frequency (ω_n) is not affected but by the addition of a zero, time response expressions are varied and rise time is reduced.

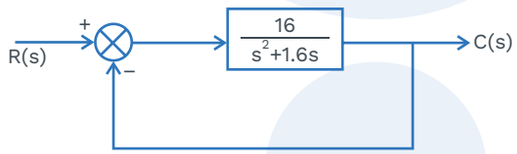
Example:

A unity feedback control system without proportional derivative controller is shown in figure. By using proportional derivative control the damping ratio is to be made 0.9. Determine the value of T_d and compare the rise time, peak time and maximum overshoot for the system.

(a) Without proportional derivative control.

(b) With proportional derivative control.

The input to the system being unit step.



Sol. The overall transfer function for the given block diagram is given by-

$$\frac{C(s)}{R(s)} = \frac{\dfrac{16}{s^2 + 1.6s}}{1 + \dfrac{16}{s^2 + 1.6s}}$$

$$\Rightarrow \quad \frac{C(s)}{R(s)} = \frac{16}{s^2 + 1.6s + 16}$$

The characteristic equation is-

$$s^2+1.6s + 16 = 0$$

The standard second order characteristic equation is given by-

$$s^2 + 2\xi\omega_n s + \omega_n^2 = 0$$

Hence,

$$\omega_n^2 = 16$$

$$\Rightarrow \quad \omega_n = 4 \text{ rad/sec}$$

$$2\xi\omega_n = 1.6$$

$$\Rightarrow \quad \xi = \frac{1.6}{2 \times 4} = 0.2$$

The damping ratio with proportional derivative controller is given by-

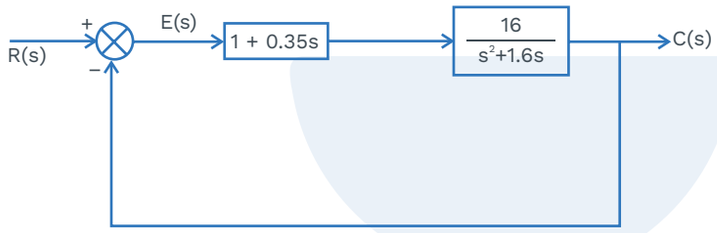
$$\xi' = \xi + \frac{\omega_n T_d}{2}$$

$\Rightarrow$ Given, $\xi' = 0.9$

$$\Rightarrow \quad 0.9 = 0.2 + \frac{4 \times T_d}{2}$$

$$\Rightarrow \quad T_d = 0.35$$

The block diagram of the given control system with proportional derivative controller is given by-



(a) Without proportional derivative controller:

Rise time,

$$t_r = \frac{\pi - \tan^{-1}\frac{\sqrt{1-\xi^2}}{\xi}}{\omega_n\sqrt{1-\xi^2}}$$

$$\Rightarrow \quad t_r = \frac{\pi - \tan^{-1}\frac{\sqrt{1-(0.2)^2}}{0.2}}{4\sqrt{1-(0.2)^2}}$$

$$\Rightarrow \quad t_r = 0.45 \text{ sec}$$

Peak time,

$$t_p = \frac{\pi}{\omega_n\sqrt{1-\xi^2}}$$

$$\Rightarrow \quad t_p = \frac{\pi}{4\sqrt{1-(0.2)^2}}$$

$$\Rightarrow \quad t_p = 0.8 \text{ sec}$$

% Maximum overshoot,

$$\% \ M_P = e^{\frac{-\xi\pi}{\sqrt{1-\xi^2}}} \times 100$$

$$\Rightarrow \quad \%M_P = e^{\frac{-0.2\pi}{\sqrt{1-(0.2)^2}}} \times 100$$

$$\Rightarrow \quad \%M_P = 52.6\%$$

(b) With proportional derivative controller, the overall transfer function is given by-

$$\frac{C(s)}{R(s)} = \frac{(1+0.35s)16}{s^2+6.4s+16}$$

Given, input is unit step i.e.

$$R(s) = \frac{1}{s}$$

$$\Rightarrow \quad C(s) = \frac{1}{s}.\frac{(1+0.35s)16}{s^2+6.4s+16}$$

$$\Rightarrow \quad C(s) = \frac{1}{s} - \frac{s+3.2}{s^2+6.4s+16} + \frac{2.4}{s^2+6.4s+16}$$

$$\Rightarrow \quad C(s) = \frac{1}{s} - \frac{s+3.2}{(s+3.2)^2+(2.4)^2} + \frac{2.4}{(s+3.2)^2+(2.4)^2}$$

By taking inverse Laplace transform-

$$c(t) = 1 - e^{-3.2t}\cos 2.4t + e^{-3.2t}\sin 2.4t$$

To determine the rise time (t_r), put c(t) = 1:

$$\Rightarrow \quad 1 = 1 - e^{-3.2t_r}\cos 2.4t_r + e^{-3.2t_r}\sin 2.4t_r$$

$$\Rightarrow \quad \sin 2.4t_r = \cos 2.4t_r$$

$$\Rightarrow \quad \tan 2.4t_r = 1$$

$$\Rightarrow \quad t_r = \frac{\tan^{-1}(1)}{2.4}$$

$$\Rightarrow \quad t_r = 0.33$$

The peak time (t_p) is determined below-

$$c(t) = 1 - e^{-3.2t}\cos 2.4t + e^{-3.2t}\sin 2.4t$$

$$\Rightarrow \quad c(t) = 1 - e^{-3.2t}[\cos 2.4t - \sin 2.4t]$$

By differentiating c(t) w.r.t. time, we get-

$$\frac{dc(t)}{dt} = 0 - e^{-3.2t}[-2.4\sin 2.4t - 2.4\cos 2.4t] - [\cos 2.4t - \sin 2.4t][-3.2e^{-3.2t}]$$

$$\Rightarrow \quad \frac{dc(t)}{dt} = e^{-3.2t}[(2.4\sin 2.4t + 2.4\cos 2.4t) + (3.2\cos 2.4t - 3.2\sin 2.4t)]$$

$$\Rightarrow \quad \frac{dc(t)}{dt} = e^{-3.2t}[-0.8\sin 2.4t + 5.6\cos 2.4t]$$

To obtain, peak time (t_p), put $\frac{dc(t)}{dt} = 0$

$$\Rightarrow \quad -e^{-3.2t}[-0.8\sin 2.4t_p + 5.6\cos 2.4t_p] = 0$$

$$\Rightarrow \quad -0.8 \sin 2.4t_p + 5.6 \cos 2.4t = 0$$

$$\Rightarrow \quad \tan 2.4t_p = 7$$

$$\Rightarrow \quad t_p = \frac{\tan^{-1}(7)}{2.4}$$

$$\Rightarrow \quad t_p = \frac{\tan^{-1}(7)}{2.4}$$

$$\Rightarrow \quad t_p = 0.604 \text{ sec}$$

The maximum value $c(t)_{max}$ is determined by putting $t = t_p$ in the time response expression-

$$c(t)_{max} = 1 - e^{-3.2\times 0.604}[\cos(2.4 \times 0.604) - \sin(2.4 \times 0.604)]$$

$$\Rightarrow \quad c(t)_{max} = 1 - 0.145[0.1208 - 0.9926]$$

$$\Rightarrow \quad c(t)_{max} = 1.1264$$

% Maximum overshoot-

$$\%M_P = \frac{[c(t)_{max} - 1] \times 100}{1}$$

$$\Rightarrow \quad \%M_P = \frac{[1.1264 - 1] \times 100}{1}$$

$$\Rightarrow \quad \%M_P = 12.64\%$$

(5) Proportional Integral (P-I) Controller

In case of proportional integral controller, the actuating signal consists of a proportional error signal added with the integral of the error signal.

i.e. $e_a(t) + K_i \int e(t)dt$

where, K_i = Integral constant

By taking Laplace transform-

$$E_a(s) = E(s) + \frac{K_i E(s)}{s}$$

The second order control system with proportional integral control action is given by-

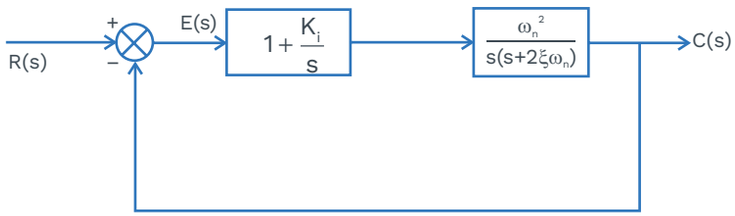


Figure-8.6. Second order system with Integral Controller

The transfer function of a closed loop second order control system using proportional integral controller is given by-

$$\frac{C(s)}{R(s)} = \frac{\left(1+\frac{K_i}{s}\right)\left[\frac{\omega_n^2}{s(s+2\xi\omega_n)}\right]}{1+\left(1+\frac{K_i}{s}\right)\left[\frac{\omega_n^2}{s(s+2\xi\omega_n)}\right]}$$

$$\Rightarrow \quad \frac{C(s)}{R(s)} = \frac{\left(s+K_i\right)\omega_n^2}{s^3+2\xi\omega_n s^2+\omega_n^2 s+K_i\omega_n^2}$$

The characteristic equation for overall transfer function is given by-

$$s^3 + 2\xi\omega_n s^2 + \omega_n{}^2 s + K_i\omega_n{}^2 = 0$$

The equation is of 3rd order, so from Routh Hurwitz Criterion we can conclude that-

- If $2\xi\omega_n < K_i$, then out of three roots two roots will have positive real value and hence, system is unstable.
- If $2\xi\omega_n > K_i$, then all the three roots have negative real part indicating a stable system.

The forward path transfer function is given by-

$$G(s) = \frac{(s+K_i)\omega_n^2}{s^2(s+2\xi\omega_n)}$$

And the feedback path transfer function is given by-

$$H(s) = 1$$

The transfer function relating E(s) and R(s) is given by-

$$\frac{E(s)}{R(s)} = \frac{1}{1+G(s)H(s)}$$

$$\Rightarrow \quad \frac{E(s)}{R(s)} = \frac{1}{1+\dfrac{(s+K_i)\omega_n^2}{s^2(s+2\xi\omega_n)}}$$

$$\Rightarrow \quad E(s) = R(s)\frac{s^2(s+2\xi\omega_n)}{s^3+2\xi\omega_n s^2+\omega_n^2 s+K_i\omega_n^2}$$

For Ramp input-

$$R(s) = \frac{1}{s^2}$$

The steady state error with proportional integral controller is zero. And without using proportional integral control, the steady state error in case of a second order control system is given by-

$$e_{ss} = 2\xi/\omega_n$$

Now, for unit parabolic input-

$$R(s) = 1/s^3$$

The steady state error without proportional integral controller is infinite and with the use of proportional integral controller, the steady state error is given by-

$$E(s) = \frac{1}{s^3}\left(\frac{s^2(s+2\xi\omega_n)}{s^3+2\xi\omega_n s^2+\omega_n^2 s+K_i\omega_n^2}\right)$$

Hence,

$$e_{ss} = \lim_{s\to 0} sE(s)$$

$$\Rightarrow \quad e_{ss} = \frac{2\xi}{K_i\omega_n}$$

Input	Steady State Error	
	Without integral Controller	With integral Controller
Unit Ramp	$\frac{2\xi}{\omega_n}$	0
Unit Parabola	∞	$\frac{2\xi}{K_i\omega_n}$

Previous Year's Questions

Introduction of integral action in the forward path of a unity feedback system result in a:
(a) Phase increase linearly with frequency
(b) amplitude remains constant
(c) amplitude increases linearly with frequency
(d) amplitude decrease linearly with frequency
Ans. (c)

(6) Proportional Integral Derivative (PID) controller-

In case of PID controller, the actuating signal consists of proportional error signal added with derivative and integral of error signal.

Hence, the actuating signal for PID controller is given by:

$$e_a(t) = e(t) + T_d \frac{de(t)}{dt} + K_i \int e(t)dt.$$

By taking the Laplace transform, we get-

$$E_a(s) = E(s) + sT_dE(s) + \frac{K_i}{s}E(s)$$

$$\Rightarrow \quad E_a(s) = E(s)\left[1 + sT_d + \frac{K_i}{s}\right]$$

The block diagram of a second order control system incorporating PID control is given below-

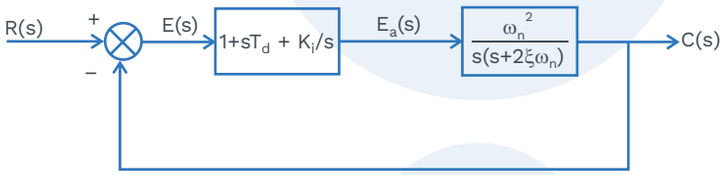


Figure-8.7. PID Controller

PID controller combines the advantages of proportional, derivative and integral controller.

Derivative Feedback Controller

In case of derivative feedback controller (or) Rate feedback (or) Tachometer feedback controller, the actuating signal is obtained as the difference between the proportional error signal and derivative of the output signal.

Hence, the actuating quantity for derivative feedback controller is given by-

$$e_a(t) = e(t) - K_t \frac{dc(t)}{dt} \text{ where, } K_t = \text{constant.}$$

By taking the Laplace transform,

$$E_a(s) = E(s) - sK_t\, C(s)$$

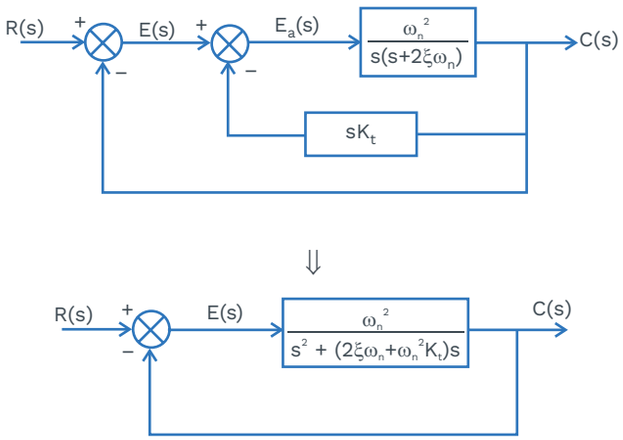


Figure-8.8. Derivative (or) Tachometer Feedback Controller

The characteristics equation of overall transfer function is given by-

$$s^2 + (2\xi\omega_n + \omega_n^2K_t)s + \omega_n^2 = 0$$

$$\Rightarrow \quad \xi' = \frac{2\xi\omega_n + \omega_n^2K_t}{2\omega_n}$$

$$\Rightarrow \quad \xi' = \xi + \frac{\omega_nK_t}{2}$$

As a result, damping ratio is increased. Hence, maximum overshoot is decreased but the rise time is increased.

The forward path transfer function G(s) and feedback path transfer function H(s) is given by-

$$G(s) = \frac{\omega_n^2}{s^2 + (2\xi\omega_n + \omega_n^2K_t)s} \text{ and } H(s) = 1$$

The transfer function relating E(s) and R(s) is given by-

$$\frac{E(s)}{R(s)} = \frac{1}{1 + G(s)H(s)}$$

$$\Rightarrow \quad \frac{E(s)}{R(s)} = \frac{1}{1 + \dfrac{\omega_n^2}{s^2 + (2\xi\omega_n + \omega_n^2K_t)s}}$$

$$E(s) = R(s)\frac{s^2 + (2\xi\omega_n + \omega_n^2K_t)s}{s^2 + (2\xi\omega_n + \omega_n^2K_t)s + \omega_n^2}$$

Previous Year's Questions

The transfer function of a PID controller is given by $G(s) = 4\left(1 + \frac{1}{2s} + 0.5s\right)$ as ω tends to infinity.

(a) Magnitude of G(jω) tends to zero and phase angle of G(jω) tends to infinity

(b) Magnitude of G(jω) tends to infinity and phase angle of G(jω) tends to zero

(c) Magnitude of G(jω) tends to infinity and phase angle of G(jω) tends to +90°

(d) Magnitude of G(jω) tends to zero and phase angle of G(jω) tends to +90°

Ans. (c)

For unit Ramp input,

$$R(s) = \frac{1}{s^2}$$

$$\Rightarrow \quad E(s) = \frac{1}{s^2}\frac{s^2 + (2\xi\omega_n + \omega_n^2 K_t)s}{s^2 + (2\xi\omega_n + \omega_n^2 K_t)s + \omega_n^2}$$

The steady state error is-

$$e_{ss} = \lim_{s\to 0} sE(s)$$

$$\Rightarrow \quad e_{ss} = \frac{2\xi}{\omega_n} + K_t$$

Example:

The overall transfer function of a control system is given by-

$$\frac{C(s)}{R(s)} = \frac{16}{s^2 + 1.6s + 16}$$

It is desired that the damping ratio be 0.8. Determine the derivative rate feedback constant K_t and compare rise time, peak time, maximum overshoot and steady state error for unit ramp input without and with derivative feedback controller.

Sol.

The characteristic equation is-

$$s^2 + 1.6s + 16 = 0$$

Comparing with the second order characteristic equation

i.e., $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$

Hence,

$$\omega_n^2 = 16$$

$$\Rightarrow \quad \omega_n = 4 \text{ rad/sec}$$

$$2\xi\omega_n = 1.6$$

$$\Rightarrow \quad \xi = 0.2$$

The damping ratio with derivative feedback controller is given by-

$$\xi' = \xi + \frac{\omega_n K_t}{2}$$

$$\Rightarrow \quad 0.8 = 0.2 + \frac{4K_t}{2}$$

$$\Rightarrow \quad K_t = 0.3$$

(a) Without derivative feedback controller-

$$t_r = \frac{\pi - \tan^{-1}\left\{\frac{\sqrt{1-\xi^2}}{\xi}\right\}}{\omega_n\sqrt{1-\xi^2}}$$

$$\Rightarrow \quad t_r = \frac{\pi - \tan^{-1}\left\{\frac{\sqrt{1-(0.2)^2}}{0.2}\right\}}{4\sqrt{1-(0.2)^2}}$$

$$\Rightarrow \quad t_r = 0.45 \text{ sec}$$

$$t_p = \frac{\pi}{\omega_n\sqrt{1-\xi^2}}$$

$$\Rightarrow \quad t_p = \frac{\pi}{4\sqrt{1-(0.2)^2}}$$

$$\Rightarrow \quad t_p = 0.8 \text{ sec}$$

% Maximum overshoot,

$$\%M_p = e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}} \times 100\%$$

$$\Rightarrow \quad \%M_p = e^{-0.64} \times 100\%$$

$$\Rightarrow \quad \%M_p = 52.6\%$$

Steady state error,

$$e_{ss} = \frac{2\xi}{\omega_n}$$

$$\Rightarrow \quad e_{ss} = \frac{2 \times 0.2}{4}$$

$$\Rightarrow \quad e_{ss} = 0.1$$

(b) With Derivative Feedback Control

The overall transfer function of the system using derivative feedback controller is given by-

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + (2\xi\omega_n + \omega_n^2 K_t)s + \omega_n^2}$$

Given: $\xi = 0.2$, $\omega_n = 4$, $K_t = 0.3$

Gray Matter Alert!!!

In case of both derivative and derivative feedback controller, the damping ratio is increased which results in decreasing maximum overshoot but the time response expression is different in both cases.

$$\Rightarrow \quad \frac{C(s)}{R(s)} = \frac{16}{s^2 + 6.4s + 16}$$

$2\xi'\omega_n = 6.4$ and $\omega_n = 4$

$$\Rightarrow \quad \xi' = \frac{6.4}{2 \times 4}$$

$$\Rightarrow \quad \xi' = 0.8$$

Rise time, $t_r = \dfrac{\pi - \tan^{-1}\left\{\dfrac{\sqrt{1-\xi^2}}{\xi}\right\}}{\omega_n\sqrt{1-\xi^2}}$

$$\Rightarrow \quad t_r = \frac{\pi - \tan^{-1}\left\{\dfrac{\sqrt{1-(0.8)^2}}{0.8}\right\}}{4\sqrt{1-(0.8)^2}}$$

$$\Rightarrow \quad t_r = \frac{\pi - \tan^{-1}(0.75)}{2.4}$$

$$\Rightarrow \quad t_r = 1.04 \text{ sec}$$

$$t_p = \frac{\pi}{\omega_n\sqrt{1-\xi^2}}$$

$$\Rightarrow \quad t_p = \frac{\pi}{\omega_n\sqrt{1-(0.8)^2}}$$

$$\Rightarrow \quad t_p = 1.3 \text{ sec}$$

% Maximum overshoot,

$$\%M_p = e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}} \times 100\%$$

$$\Rightarrow \quad \%M_p = e^{-\frac{0.8\pi}{\sqrt{1-(0.8)^2}}} \times 100\%$$

$$\Rightarrow \quad \%M_P = 1.52\%$$

Steady state error,

$$e_{ss} = \frac{2\xi}{\omega_n} + K_t$$

$$\Rightarrow \quad e_{ss} = \frac{2 \times 0.2}{4} + 0.3$$

$$\Rightarrow \quad e_{ss} = 0.4$$

Previous Year's Questions

The transfer function of a phase lead compensator is given by:

$$D(s) = \frac{3\left(s + \frac{1}{3T}\right)}{\left(s + \frac{1}{T}\right)}$$

The frequency (in rad/sec), at which $\angle D(j\omega)$ is maximum is:

(a) $\sqrt{\dfrac{3}{T^2}}$ (b) $\sqrt{3T}$

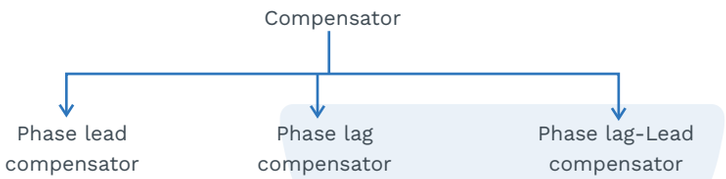
(c) $\sqrt{\dfrac{1}{3T^2}}$ (d) $\sqrt{3T^2}$

Ans. (c)

COMPENSATORS

If the performance of a control system is not upto expectations as per desired specifications, then some changes in the system are needed to obtain the desired performance. The change can be in the form of adjustment of forward path gain (or) inserting a compensating device in control system.

The forward path gain adjustment method is not suitable wherein the system becomes unstable due to such adjustment and hence, compensation network is introduced in the system.



(1) Phase Lead Compensator

The figure 8.9 represents a phase lead compensation network wherein the phase of the output voltage leads the phase of the input voltage for sinusoidal inputs.

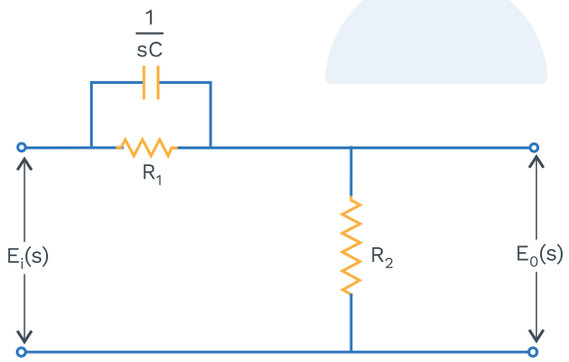


Figure-8.9. Phase-Lead Compensator Network

The transfer function of a phase lead network is given by-

$$\frac{E_0(s)}{E_i(s)} = \frac{\alpha(1+sT)}{(1+s\alpha T)}$$

where $\alpha < 1$,

i.e. $\alpha = \frac{R_2}{R_1 + R_2}$ and $T = R_1C$

The transfer function expressed in sinusoidal form is given by-

$$\frac{E_0(j\omega)}{E_i(j\omega)} = \frac{\alpha(1+j\omega T)}{(1+j\omega\alpha T)}$$

The pole-zero configuration of phase-lead compensator is given by-

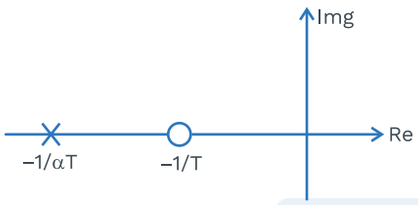


Figure-8.10. Pole zero configuration of Phase lead compensator

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The steady state error in a control system can be reduced by increasing forward path gain but on the other hand this increase in forward – path gain results in making the system more oscillatory (or) sometimes unstable.

The gain adjustment improves the steady state accuracy of the system at the cost of driving the system toward instability.

The bode plot for transfer function of phase-lead compensator-

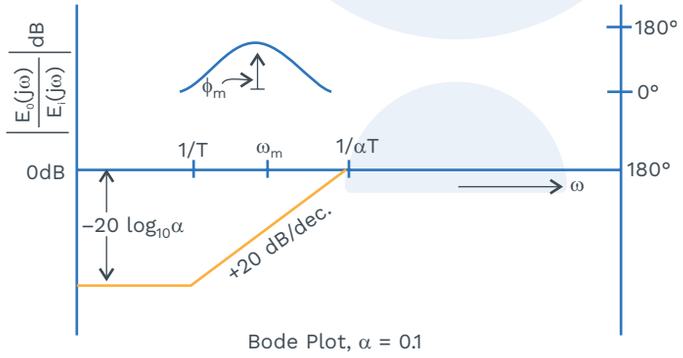


Figure-8.11. Bode Plot for Phase Lead Network

The two corner frequencies are-

$\omega_1 = \frac{1}{T}$: Lower corner frequency

$\omega_2 = \frac{1}{\alpha T}$: Upper corner frequency

The maximum phase lead occurs at mid frequency (ω_m) between upper and lower corner frequencies.

i.e $\log_{10}\omega_m = \frac{1}{2}\left[\log_{10}\left(\frac{1}{T}\right) + \log_{10}\left(\frac{1}{\alpha T}\right)\right]$

$$\Rightarrow \qquad \omega_m = \frac{1}{\sqrt{\alpha}T}$$

The phase angle $\angle E_0(j\omega) / E_i(j\omega)$ can be calculated as-

$$\angle \frac{E_0(j\omega)}{E_i(j\omega)} = \tan^{-1}(\omega T) - \tan^{-1}(\omega \alpha T)$$

At $\omega = \omega_m = \frac{1}{\sqrt{\alpha}T}$, the phase angle ($\phi_m$) is-

$$\phi_m = \tan^{-1}\left(\frac{1}{\sqrt{\alpha}\,T}.T\right) - \tan^{-1}\left(\frac{1}{\sqrt{\alpha}T}.\alpha T\right)$$

$$\Rightarrow \qquad \phi_m = \tan^{-1}\left(\frac{1}{\sqrt{\alpha}}\right) - \tan^{-1}\left(\sqrt{\alpha}\right)$$

$$\Rightarrow \qquad \tan\phi_m = \frac{\frac{1}{\sqrt{\alpha}} - \sqrt{\alpha}}{1 + \frac{1}{\sqrt{\alpha}}.\sqrt{\alpha}}$$

$$\sin\phi_m = \frac{1-\alpha}{1+\alpha}$$

$$\cos\phi_m = \frac{2\sqrt{\alpha}}{1+\alpha}$$

Previous Year's Questions

The transfer function C(s) of a compensator is given below:

$$C(s) = \frac{\left(1+\frac{s}{0.1}\right)\left(1+\frac{s}{100}\right)}{(1+s)\left(1+\frac{s}{10}\right)}$$

The Frequency range in which the phase (lead) introduced by the compensator reaches the maximum is-

(a) $0.1 < \omega < 1$ (b) $1 < \omega < 10$
(c) $10 < \omega < 100$ (d) $\omega > 100$

Ans. (a)

- According to the bode plot, the lead network allows to pass high frequencies and low frequencies are attenuated.
- As the gain is reduced at low frequencies, additional gain is needed in the system to account for the reduction in gain.

(2) Phase Lag Compensator

In case of phase-lag compensators, the phase of output voltage lags the phase of input voltage for sinusoidal inputs-

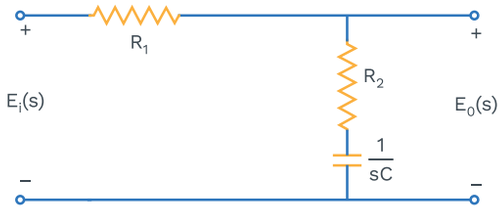


Figure-8.12. Phase Lag Compensator

The transfer function of phase lag compensator is given by-

$$\frac{E_0(s)}{E_i(s)} = \frac{1+sT}{1+s\beta T}$$

where, $\beta > 1$

i.e. $$\beta = \frac{R_1 + R_2}{R_2}$$

$$T = R_2C$$

The transfer function can be expressed in sinusoidal form as-

$$\frac{E_0(j\omega)}{E_i(j\omega)} = \frac{1+j\omega T}{1+j\omega\beta T}$$

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For phase lead compensator, zero is nearer to the origin as compared to pole and hence the effect of zero is dominant, therefore, whenever phase lead compensator is introduced in cascade with forward path of transfer function, phase shift is increased.

The pole zero configuration and bode plot for transfer function of a phase lag compensator is as shown in the figure below.

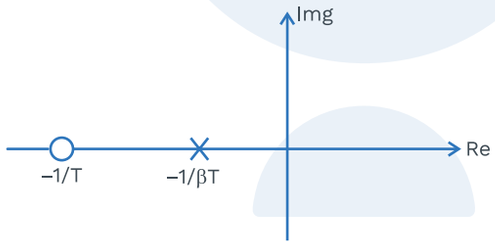


Figure-8.13. Pole Zero Configuration of Phase Lag Network

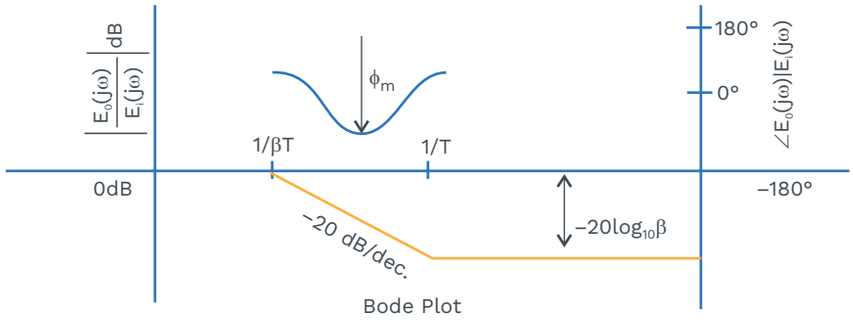


Figure-8.14. Bode Plot for Phase Lag Compensator

The two corner frequencies are given by-

$\omega_1 = \frac{1}{T}$ i.e upper corner frequency.

$\omega_2 = \frac{1}{\beta T}$ i.e. lower corner frequency.

The maximum phase lag ϕ_m occurs at mid frequency (ω_m) between upper and lower corner frequencies. Hence,

$$\log_{10}\omega_m = \frac{1}{2}\left[\log_{10}\left(\frac{1}{\beta T}\right) + \log_{10}\left(\frac{1}{T}\right)\right]$$

$$\Rightarrow \quad \omega_m = \frac{1}{\sqrt{\beta}\,T}$$

The phase angle $\angle E_0(j\omega)/E_i(j\omega)$ can be calculated as-

$$\angle\frac{E_0(j\omega)}{E_i(j\omega)} = \tan^{-1}(\omega T) - \tan^{-1}(\omega\beta T)$$

At $\omega = \omega_m = \frac{1}{\sqrt{\beta}\,T}$, hence, the phase angle (ϕ_m) is

$$\tan\phi_m = \frac{1-\beta}{2\sqrt{\beta}}$$

$$\sin\phi_m = \frac{1-\beta}{1+\beta}$$

As $\beta > 1$, implies $\frac{1-\beta}{1+\beta}$ is negative resulting in negative ϕ_m which means phase angle is lagging.

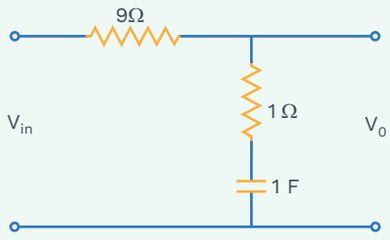
According to bode plot, the phase lag compensator allows to pass low frequencies and attenuates high frequencies.

(3) Phase Lag Lead compensator:

- The phase lead compensator shifts the gain crossover frequency point to a higher value and therefore, the bandwidth is increased thus improving the speed of response and overshoot is reduced but the steady state error does not show much improvement.
- The phase lag compensator shifts the gain crossover frequency point to a lower value thus decreasing the bandwidth and improvement in steady state error but the speed of response is reduced.
- The speed of response and steady state error can be improved simultaneously, if both phase lag and phase lead compensators are used.
- Hence, a single-phase lag lead compensator is used which combines the properties of both lead and lag compensators.

Previous Year's Questions

For the network shown in the figure below, the frequency (in rad/s) at which the maximum phase lag occurs is_____.



Ans. **(0.316)**

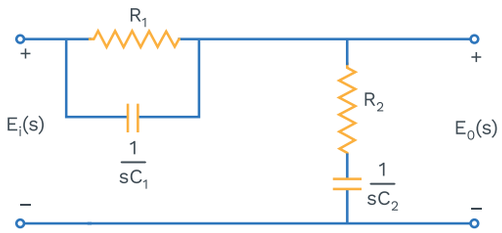


Figure-8.15. Phase lag lead compensator.

The transfer function of phase lag lead compensator is given by-

$$\frac{E_0(s)}{E_i(s)} = \frac{(1+sT_1)}{(1+s\alpha T_1)}\frac{(1+sT_2)}{(1+s\beta T_2)}$$

where,

$T_1 = R_1C_1$, $T_2 = R_2C_2$

$\alpha < 1$, $\beta > 1$ and $\alpha\beta = 1$

The pole zero configuration for phase lag lead compensator is given by-

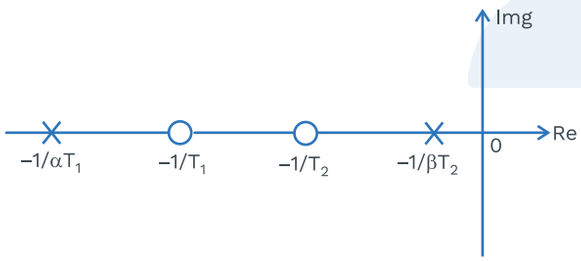


Figure-8.16. Pole zero configuration of Phase Lag Lead compensator

The bode plot for transfer function of phase lag lead compensator is as shown-

Gray Matter Alert!!!

For phase lag compensator, the pole is nearer to the origin as compared to zero. Hence, the pole effect is dominant. Therefore, the phase lag network when introduced in cascade with forward path of transfer function, the phase shift is reduced.

Previous Year's Questions

The transfer function of two compensator are given below:

$$C_1 = \frac{10(s+1)}{(s+10)}, C_2 = \frac{s+10}{10(s+1)}$$

Which one of the following statements is correct?

(a) C_1 is lead compensator and C_2 is a lag compensator

(b) C_1 is a lag compensator and C_2 is a lead compensator.

(c) Both C_1 and C_2 are lead compensator

(d) Both C_1 and C_2 are lag compensator

Ans. (a)

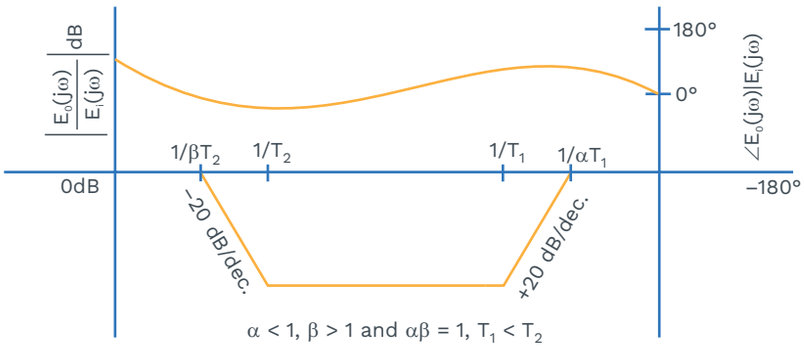


Figure-8.17. Bode Plot for phase Lag Lead compensator.

The transfer function expressed is sinusoidal form is given by-

$$\frac{E_0(j\omega)}{E_i(j\omega)} = \frac{(1+j\omega T_1)}{(1+j\omega\alpha T_1)}\frac{(1+j\omega T_2)}{(1+j\omega\beta T_2)}$$

Previous Year's Questions

The system 900/(s+1) (s+9) is to be compensated such that its gain crossover frequency becomes same as its uncompensated phase crossover frequency and provides a 45° phase margin. To achieve this, one may use:

(a) a lag compensator that provides an attenuation of 20 dB and a phase lag of 45° at the frequency of $3\sqrt{3}$ rad / s .

(b) a lead compensator that provides an amplification of 20 dB and a phase lead of 45° at the frequency of 3 rad/s.

(c) a lag-lead compensator that provides an amplification of 20 dB and a phase lag of 45° at the frequency of $\sqrt{3}$ rad/s.

(d) a lag-lead compensator that provides an attenuation of 20 dB and phase lead of 45° at the frequency of 3 rad/s.

Ans. (d)

PRACTICE QUESTIONS

Q1 **A temperature control system is usually very sluggish. In order to improve its dynamics-**
(a) a PI controller can be used
(b) an I controller can be used
(c) a PID controller with large I and a negligible D action can be used.
(d) a PD controller can be used.

Sol: **Ans. (d)**

Q2 **A plant is controlled by a proportional controller. If a time delay element is introduced in the loop then-**
(a) gain margin remains the same **(b) phase margin remains the same**
(c) gain margin decreases **(d) phase margin decreases**

Sol: **Ans. (d)**

Q3 **The transfer function of a basic PD controller is given by (Here, K are real constants)-**
(a) $K_0 + \frac{K_1}{s} + K_2 s$ **(b)** $K_0 + K_2 s$
(c) $K_1 s + K_2 s$ **(d)** $K_0 + \frac{K_1}{s}$

Sol: **Ans. (b)**

Q4 **A first order system with a proportional controller in the negative feedback loop has an offset to a step input. This offset can be eliminated by-**
(a) adding a derivative mode to the controller
(b) adding an integral mode to the controller
(c) decreasing the magnitude of the gain of the proportional controller
(d) adding a delay in the controller loop

Sol: **Ans. (b)**

Q5 **A second order feedback system is found to be oscillating with a high frequency. The oscillation-**
(a) can be reduced by increasing the proportional action.
(b) can be reduced by increasing the integral action.
(c) can be reduced by increasing the derivative action.
(d) cannot be reduced.

Sol: **Ans. (c)**

Q6 **An integral control is used to-**
(a) improve the transient response **(b) reduce the offset**
(c) eliminate the offset **(d) reduce the setting time**

Sol: **Ans. (c)**

Q7 A PID controller has the transfer function $2+\frac{0.4}{s}$ with the unit of time expressed in minutes. The parameters proportional band and reset time for the above controller are respectively-

(a) 200% and 0.4 min
(b) 50% and 0.4 min
(c) 200% and 5 min
(d) 50% and 5 min

Sol: Ans. (d)

Q8 The proportional band of a PI controller is 50% and the reset time is 0.2 s. The transfer function of the controller is given by-

(a) $50\left(1+\frac{1}{0.2s}\right)$
(b) $50\left(1+\frac{1}{5s}\right)$
(c) $2\left(1+\frac{1}{0.2s}\right)$
(d) $2\left(1+\frac{1}{s}\right)$

Sol: Ans. (c)

Q9 For a position servo system, the open loop transfer function is $G(s)=\frac{1}{s(s+1)}$. A first order compensator is designed in a unity feedback configuration such that the poles of the compensated system are placed at $-1 \pm j1$ and -4. Then the transfer function of the compensated system is-

(a) $\frac{s+3}{2(s+3)}$
(b) $\frac{2s+3}{s+5}$
(c) $\frac{5(s+1.6)}{s+5}$
(d) $\frac{3(2s+3)}{s+4}$

Sol: Ans. (c)